

Six circles in a rectangle: a proof by pure geometry

Figure 1 shows a rectangle and six circles. Wherever two of them look tangent, they are tangent. The red segment joins the centres of the two largest circles, and it appears to be exactly as long as the rectangle is high. Dan asked for a proof on Mathematics Stack Exchange [1]. The answers there, his own among them, used coordinates, polynomials or trigonometry, so he asked again on MathOverflow for an argument using only pure geometry [2]. This note gives one.

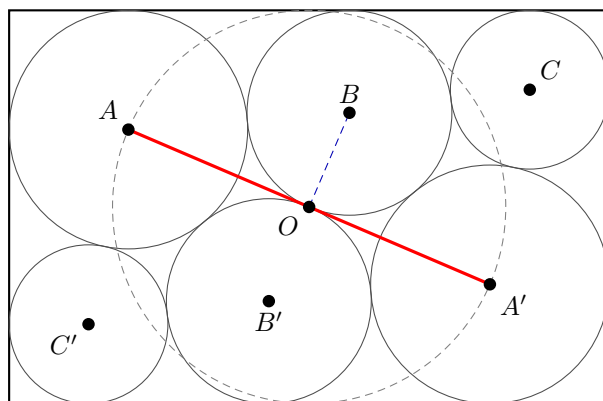


Figure 1: The top three circles touch the top side, A and C also touch the left and right sides, and B passes through the centre O . The bottom three are the images of the top three under the half-turn about O . The dashed circle has centre O and touches the top and bottom sides. The claim is that the red segment AA' equals the height of the rectangle, or equivalently that A lies on the dashed circle.

Let the rectangle have centre O and half-height h . Write A, B, C for the centres of the top three circles, from left to right, and a, b, c for their radii. Besides the contacts with the sides, the tangencies are: B touches A, C and A' , and C touches A' . With $h = 1$ one can compute $a \approx 0.6066$, $b \approx 0.5210$ and $c \approx 0.4037$; the figures are drawn from this solution, though the proof never uses the numbers. We use only one feature of the picture that the tangencies do not spell out: each radius is less than h , so every centre in the top row lies above the horizontal through O .

Before reading on, try the first step. Why is it enough to show that $OA = h$?

The reduction. Circle B touches the congruent circles A and A' , so $BA = BA'$. Since O is the midpoint of AA' , the line OB is the median of an isosceles triangle and so is perpendicular to AA' . The red segment is $2OA$, and it equals the height $2h$ exactly when $OA = h$. Call Ω the circle with centre O and radius h , dashed in Figure 1. It touches the top side at its midpoint U , and the claim is that A lies on Ω .

The plan. Let E' be the left end of the horizontal diameter of Ω . If we knew that the line AE' ran in a known direction, the triangle $OE'A$ would have one side $OE' = h$ and two known angles, and we could hope to show it is isosceles. The direction comes from the line OB_t , where B_t is the point at which circle B touches the top side. Steps 1 and 2 show that the line of centres AC' is parallel to OB_t . Step 3 shows that this line passes through E' , and Step 4 finishes.

Put $\psi = \angle UOB_t$. In triangle OB_tB the sides BO and BB_t are radii of circle B , and BB_t is

parallel to OU , so

$$\angle UOB = 2\psi,$$

twice the base angle. The point B_t lies to the right of U : otherwise A and B would both lie above the horizontal through O and on or to the left of OU , and the angle AOB would be acute, contradicting the reduction.

1. A bisector

One fact about tangent lengths does most of the work. Two circles of radii r and s that touch each other, and touch a line on the same side, touch the line at points $2\sqrt{rs}$ apart. (Their centres are $r + s$ apart and differ in height by $|r - s|$, and $(r + s)^2 - (r - s)^2 = 4rs$.) For radii 1 and 4 the contact points are 4 apart.

On the left side, A and C' touch each other and the side, at heights $h - a$ and $-(h - c)$, so $2h - a - c = 2\sqrt{ac}$. On the top side, let A_t and C_t be the contact points of A and C , so that $A_tC_t = 2\sqrt{ab} + 2\sqrt{bc}$. In short,

$$\sqrt{a} + \sqrt{c} = \sqrt{2h}, \quad A_tC_t = 2\sqrt{b}(\sqrt{a} + \sqrt{c}). \quad (1)$$

The first ties the two left-hand radii to the height. The second measures the top side between the corner contacts.

Circle B passes through O and touches the top side at B_t , so B_t is one end of a vertical diameter of length $2b$. In the right-angled triangle formed by this diameter and O , the leg OB_t satisfies $OB_t^2 = 2b \cdot h$, the diameter times the projection of the leg onto it. With (1),

$$A_tC_t = 2\sqrt{2bh} = 2OB_t.$$

Let M be the midpoint of A_tC_t . Along the top side, M lies $\frac{1}{2}(a - c)$ to the right of U , because A_t and C_t are a and c in from the corners. And B_t lies $2\sqrt{ab} - \sqrt{b}(\sqrt{a} + \sqrt{c}) = \sqrt{b}(\sqrt{a} - \sqrt{c})$ to the right of M . Both displacements have the sign of $a - c$, and B_t is to the right of U , so $a > c$ and the order along the top is U, M, B_t . The ratio in which M divides UB_t is

$$\frac{UM}{MB_t} = \frac{\frac{1}{2}(a - c)}{\sqrt{b}(\sqrt{a} - \sqrt{c})} = \frac{\sqrt{a} + \sqrt{c}}{2\sqrt{b}} = \frac{h}{\sqrt{2bh}} = \frac{OU}{OB_t},$$

using $a - c = (\sqrt{a} - \sqrt{c})(\sqrt{a} + \sqrt{c})$ and then (1). By the converse of the angle bisector theorem in triangle UOB_t , the line OM bisects the angle UOB_t .

2. A parallel

Step 1 gave a direction at O . To carry it to the line of centres AC' we need a line through both circles A and C' whose direction we can read off, and a homothety between the two circles supplies one. Let T be the point where circles A and C' touch, and let C'_b be the lowest point of circle C' . The homothety with centre T and negative ratio that maps circle A to circle C' maps the highest point A_t of the first to the lowest point C'_b of the second, so A_t, T and C'_b are collinear. From A_t to C'_b the horizontal and vertical separations are $a - c$ and $2h$, exactly twice those from O to M . Hence A_tT is parallel to OM and makes the angle $\psi/2$ with the vertical. The triangle AA_tT is isosceles with base angles $\psi/2$, so its angle at A is $180^\circ - \psi$ and the radius AT makes the angle ψ with the downward vertical, and so does the line of centres AC' , which passes through T . Both AC' and OB_t rise to the right, so

$$AC' \parallel OB_t.$$

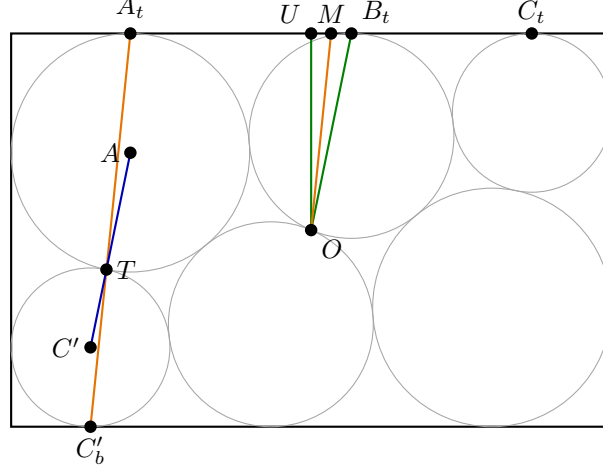


Figure 2: Steps 1 and 2. The orange line OM bisects the angle between the green lines OU and OB_t . The orange line from A_t through the contact point T to C'_b is parallel to it, and the blue line of centres AC' is then parallel to OB_t .

3. The line of centres passes through E'

We now need to locate AC' , not just its direction, and the length to aim for is h . The circle Ω puts that length on the line OB_t . Let K be the point where OB_t crosses Ω , so $OK = h$. It lies between O and B_t , because OB_t is the hypotenuse of the right-angled triangle OUB_t and so is longer than $OU = h$. The tangents to Ω at U and K meet at a point on the bisector of the angle UOK , which is OM ; the tangent at U is the top side, so they meet at M . Hence $MK = MU$ and $MK \perp OB_t$ (Figure 3).

Now let S be the midpoint of AC' and S_0 the foot of the perpendicular from S to the horizontal through O . The centres A and C' are a and c in from the left side, and the half-width of the rectangle is $\frac{1}{2}(A_t C_t + a + c)$, so

$$OS_0 = \frac{1}{2}A_t C_t = OB_t, \quad SS_0 = \frac{1}{2}(a - c) = UM,$$

with S_0 to the left of O and S below it. In words: S_0 is as far from O as B_t is, and S sits as far below the horizontal as M sits to the right of U .

Let the line AC' cross the horizontal through O at X . Compare the right-angled triangles MKB_t and SS_0X , shaded in Figure 3. Their legs MK and SS_0 both equal UM . The angle at M is ψ , because the angle at B_t is $\angle OB_t U = 90^\circ - \psi$. The angle at S is also ψ , because SX lies along AC' , which is parallel to OB_t . So the triangles are congruent, and

$$S_0 X = KB_t = OB_t - h.$$

Going up AC' from S to the horizontal moves to the right, so X lies to the right of S_0 , at distance $OS_0 - S_0 X = h$ from O . That is, $X = E'$.

4. The isosceles triangle

In triangle $OE'A$ the side $E'A$ lies along AC' , which is parallel to OB_t and so makes the angle $90^\circ - \psi$ with the horizontal $E'O$. Turning the figure through a right angle about O takes the upward vertical to the leftward horizontal and, since $OA \perp OB$, takes the line OB to the line OA . So the angle $E'OA$ equals the angle $UOB = 2\psi$. The three angles are

$$\angle E'OA = 2\psi, \quad \angle OE'A = 90^\circ - \psi, \quad \angle OAE' = 180^\circ - 2\psi - (90^\circ - \psi) = 90^\circ - \psi.$$

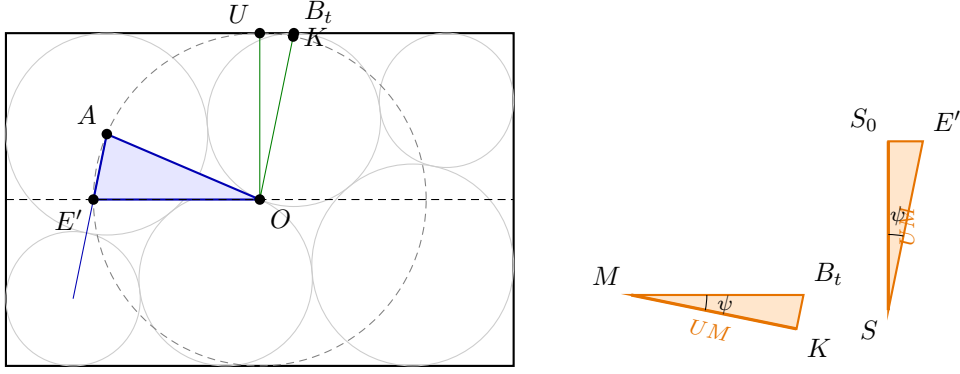


Figure 3: Steps 3 and 4. Left: the line of centres AC' passes through E' , the left end of the horizontal diameter of Ω , and the blue triangle $OE'A$ is isosceles. Right: the two small right triangles MKB_t and SS_0E' , magnified 22 times at the same scale. Their thick legs both equal UM and the marked angles are both ψ , so they are congruent.

The angles at E' and A are equal, so $OA = OE' = h$. The point A lies on Ω , and the red segment AA' equals the height $2h$. $\square$

Acknowledgements. The problem and its history are due to Dan [1, 2]. A longer version of this proof first appeared as the author's answer on MathOverflow [3]. The author used AI tools in developing this proof and takes full responsibility for its content.

References

- [1] Dan, *Six circles in a rectangle: show that two lengths are equal*, Mathematics Stack Exchange question 5112314 (4 December 2025), <https://math.stackexchange.com/q/5112314>.
- [2] Dan, *Six circles in a rectangle: show that two lengths are equal, using pure geometry*, MathOverflow question 515498 (24 September 2026), <https://mathoverflow.net/q/515498>.
- [3] V. Jandhyala, answer to MathOverflow question 515498 (1 October 2026), <https://mathoverflow.net/a/515659>.

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